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EXTERNAL GPU

UNFINISHED LECTURE NOTES

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These notes are, by their very nature, unfinished, and they improve with every reader. If you spot an error, disagree with a framing, or want to add a source, a question, or a position, open a pull request at https://github.com/Quillstacks/LectureMaterial/tree/main/lecturenotes/notes_egpu. Every contribution is welcome.



*“ONCE MEN TURNED THEIR THINKING OVER TO MACHINES IN THE HOPE
THAT THIS WOULD SET THEM FREE.”*

REVEREND MOTHER GAIUS HELEN MOHIAM, IN FRANK HERBERT, DUNE

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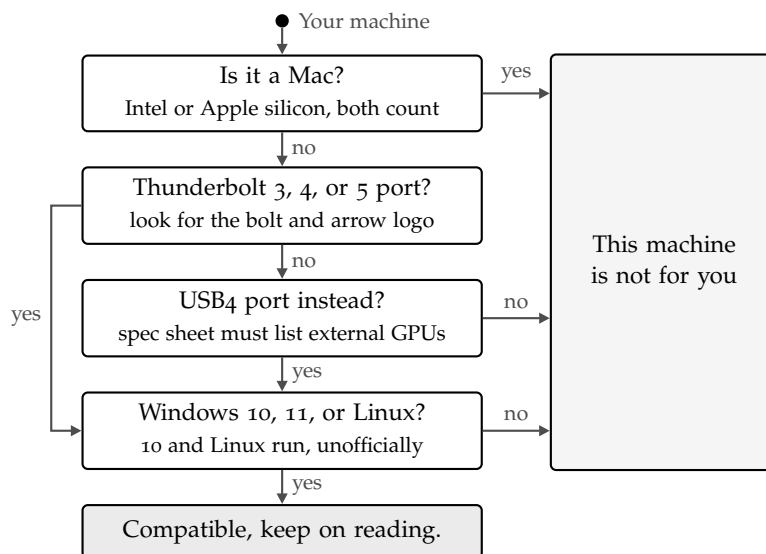
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Introduction

THIS IS A MANUAL. It gets a desktop NVIDIA graphics card with 16 GB of memory running on your laptop through the lab's Sonnet eGPU Breakaway Box 750ex: A Thunderbolt enclosure with one PCIe slot, a 750 W power supply, four USB-A ports, and Gigabit Ethernet. Most of the work happens before you plug anything in, because most laptops cannot drive it. Check yours first. If a step in this manual is deprecated or you found a better way, contribute it back. Of course feel free to add any information that you think is useful for the next person who wants to use the eGPU.

THE OFFICIAL DOCUMENTATION. Sonnet's (no, not your Claude Sonnet) product page carries the tech specs, the quick start guide, and the macOS and Windows graphics guides.



POWERFUL, BUT it is quite picky. Last time we looked these were the compatibility specs. Feel free to challenge and adjust the document if needed.

Quickstart

2026-10-01 · local draft

Aaaah-ah-ah-ah, aaaah-ah-ah-ah, Thunder!

LUCKY YOU. Your machine is compatible. Ask for the box next. You get a time slot and hand the box back with its cable and power cord. Never attempt to open the case, and don't drop it.

PREPARE THE LAPTOP BEFORE PICKING UP THE BOX. I want to see it running before you will be allowed to take it with you. On Windows, install pending updates, the newest BIOS from your laptop's manufacturer, and the NVIDIA driver from NVIDIA's site; outdated firmware can hide the card ¹. On Ubuntu, run `sudo ubuntu-drivers install`.

PLUG IN. Use the cable that comes with the box, connect the power cord, and flip the switch on the back. The front LED lights once the laptop is on. Keep the ventilator shafts clear.

APPROVE THE DEVICE. Windows on Thunderbolt 3 asks once, choose Always connect. On Ubuntu, run `boltctl enroll` with the device id that `boltctl list` shows.

CHECK THE CARD. `nvidia-smi` in a terminal shows it with its 16 GB.

ALWAYS EJECT BEFORE YOU UNPLUG. Stop every process `nvidia-smi` lists, then on Windows use the NVIDIA eject icon in the taskbar tray. Pulling the cable from a running card can crash the laptop.

THE SOUNDTRACK for first use. Meant to be played loud. Your neighbours will understand.



SCAN THE CODE and send a mail stating your name, course, laptop model, why and what you want to run.



¹ Sonnet Technologies. egpu breakaway box graphics guide for windows 10 and 11. https://www.sonnettech.com/support/downloads/manuals/windows_egpu_info.pdf, 2023

CHECKS. Nothings in Device Manager means the driver is missing. Nothing at all means the box is not detected: Check cable, power switch, and port.

AFTER A RECONNECT on Windows, click Connect GPU in the same tray icon.

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